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SWE 3409 · MOBILE APPLICATION DEVELOPMENT · INDIVIDUAL ASSIGNMENT 2

UI and UX design

Document: docs/MOB_A2_2528255_UIUX.pdf · 12 October 2026 · Pages 1–6

> **My name is Teta Erssie.**
> This is the first sentence on the Home screen and the first line of README.md.

1. Identity and concept

Item	Value
Student	Teta Erssie
Registration number	2528255
Programme	BSc Software Engineering, INES-Ruhengeri, Musanze
App name	TetaPortfolio
Verification code	MOB-A2-8255 (last four digits of the registration number)
Android package	rw.ac.ines.ug2528255.tetaportfolio
Version	1.0.0

Intended recruiter

The app is for a short internship or first-job meeting. A recruiter installs the APK, opens it without Expo Go, and should understand who I am within two minutes, including when the phone is offline. The Home screen leads with the name, a headline, a portrait, availability, a short biography and the verification code. Work and education are one tap away.

Visual direction

A light, portrait-first layout. The background is a soft teal–lavender–pink wash. Content sits on frosted white cards with rounded corners. Text is Plus Jakarta Sans in deep ink, so the page stays readable on a narrow phone. The portrait is an illustrated avatar until a photo is taken or chosen.

Teal
Calm, approachable

Lavender
Primary actions

Pink
Warm accent

White cards
Reading surface

2. Information architecture

Five bottom tabs. The Work tab contains a nested stack: the project list stays mounted, and a project opens a detail screen. Android Back returns to the list and keeps the session.

Launch Welcome, then Continue → **Home** Name, portrait, bio, code → **About me** Hello, Skills, Academic → **Work** Project cards → **Details** Stack screen, Back

Edit Headline, bio, skill, availability, photo → **Validate** Field messages. Invalid save is blocked. → **Save** AsyncStorage on this device only → **Info** Identity, version, data-use, links

Tab label	Assignment section	Route
Home	Introduction	src/app/(tabs)/index.tsx
About me	Skills and education	src/app/(tabs)/skills.tsx
Work	Project portfolio	src/app/(tabs)/projects/index.tsx
Work → detail	Project Details stack	src/app/(tabs)/projects/[id].tsx
Edit	Profile editor and media	src/app/(tabs)/profile.tsx
Info	About and release	src/app/(tabs)/about.tsx

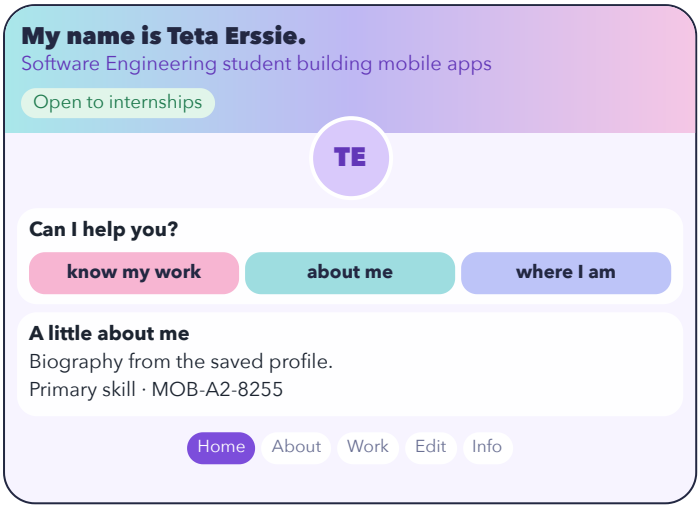
Profile editing and media

1. Edit opens a controlled form loaded from the saved profile.
2. Camera or Gallery asks permission. Denial keeps the current photo and explains how to open Settings. Cancelling the picker leaves the photo unchanged.
3. A chosen image is previewed at once. Replace and Remove are available. Remove asks for confirmation, then the initials avatar returns. Save stores the change.

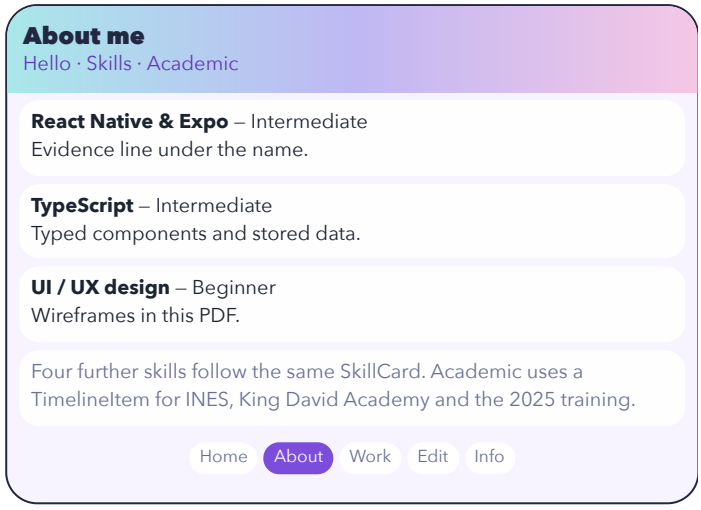
Saved-state flow

Save writes the headline, biography, primary skill, availability and local photo reference with AsyncStorage (key `@tetaportfolio/profile`). Closing and reopening restores that profile. Reset saved profile deletes it and restores the defaults. If the stored value cannot be read, Home shows a warning and the default profile. GitHub and LinkedIn are the only features that need a connection, and both are labelled as such. The rest of the portfolio works in Airplane Mode.

3. Home and skills

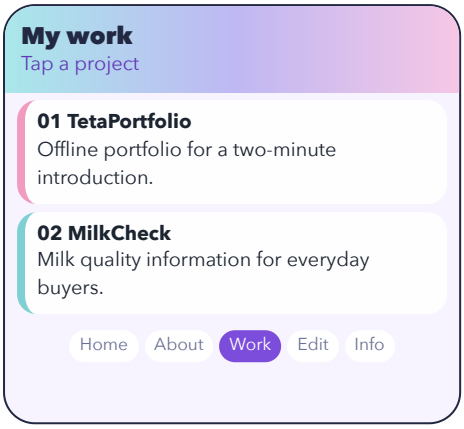


1 Required sentence is the page title. **2** Headline and availability come from the saved profile. **3** Portrait or initials. **4** Three shortcuts. Contact me is labelled as needing internet. **5** Biography, primary skill and verification code.

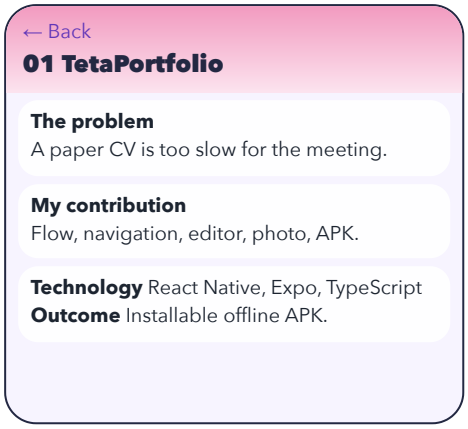


6 Segmented control switches Hello, Skills and Academic without leaving the tab. **7** Each skill is a reusable SkillCard: name, level, evidence. **8** Education is a reusable TimelineItem: period, title, place, detail. Seven skills and three education entries are bundled.

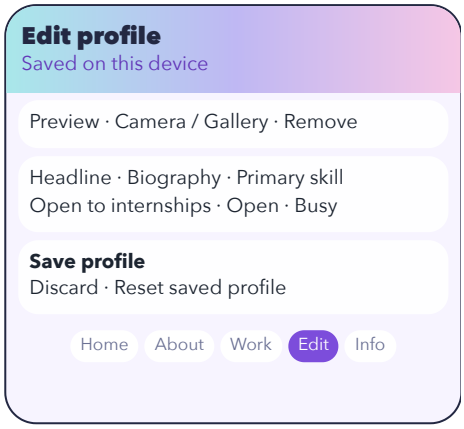
4. Work, project details and profile editor



9 ProjectCard shows number, title and summary, and opens the stack.



10 Unknown ids show "We couldn't find that project" and a Back action.



11 Messages sit under the field that failed. Save stays blocked.

Required error and empty states

State	What the screen shows
Headline	Required. 10–80 characters.
Biography	Required. 40–300 characters. Links and email addresses are rejected.
Primary skill	Required. 2–40 characters. Letters, numbers and + # . - / & only.
Availability	"Choose your availability status." One of three options must be selected.
Invalid save	"Fix the highlighted fields before saving." Nothing is written.
Permission denied	"Camera permission was denied. Your current photo was kept." Settings can be opened.
Picker cancelled	"No photo selected. Nothing was changed."
Empty lists	Dotted empty state if skills, education or projects are absent: "No skills listed yet", "No education entries yet", "No projects yet".
Storage error	Home warns that the saved profile could not be read and shows the default.

5. Colour

Tokens live in `src/design-system/colors.ts`. Contrast below is the WCAG ratio of the named pair. Body text is at least 4.5:1.

Role	Colour	Pair	Ratio
Primary text	#232842	on white	14.5:1
Secondary text	#454B6B	on white	8.5:1
Muted labels	#6B7194	on white	4.8:1
Primary button	#7C4DDB	white label	5.3:1
Name accent	#6237B8	on white	7.6:1
Error	#B42318	on white / #FDEBEA	6.6:1
Success	#1E7A4E	on white	5.3:1

Accent tiles (pink, teal, periwinkle, lavender) use a dark text colour on a pale fill for the words, and a stronger fill only behind large decorative areas. The reading text stays ink, not white-on-pastel.

Typography, spacing and touch

Decision	Choice
Typeface	Plus Jakarta Sans. Titles ExtraBold, labels Bold, body Regular. Loaded with expo-font.
Home title	28px, line height 36. Section titles about 18–26px. Body 15px with line height 22–24.
Spacing	4, 8, 12, 16, 24, 32. Card radius 16–24. Screen padding stays inside the safe area.
Touch	Buttons and availability choices use a 48px minimum height (Android guidance).
Tabs	Floating dock. The active tab shows its label. Inactive tabs keep an accessibility label.

Accessible labels and large text

- The name uses `accessibilityRole="header"`. Buttons, tabs, radios, the back control and links expose labels. The verification code is read character by character.
- Skill cards, timeline rows, project cards and info rows expose a full label so a screen reader does not have to guess from the layout.
- Text scaling stays on for body copy. Rows wrap instead of using a fixed height. The tab label and the welcome wordmark cap their multiplier (1.2 and off for the decorative wordmark only) so the dock and the launch screen do not overflow. The welcome screen is dismissed with Continue before the portfolio is read.
- One light theme. There is no dark theme in this release.

6. Wireframes and the built screens

The wireframes on pages 3 and 4 use the same labels, order and states as the Expo Router screens. The table is the check that the built app still matches this document.

Wireframe	Built screen	Match
Home title sentence	index.tsx	Exact sentence “My name is Teta Erssie.”
Headline, badge, portrait, bio, code	index.tsx, HeroPortrait, VerificationCode	Same order. Values come from the saved profile.
About me segments	skills.tsx	Hello, Skills, Academic. SkillCard and TimelineItem.
Work cards	projects/index.tsx	TetaPortfolio and MilkCheck. Empty state if the list is empty.
Detail stack	projects/[id].tsx	Problem, contribution, technology, outcome, Back.
Editor and photo	profile.tsx	Four fields, three availability choices, camera, gallery, replace, remove.
Info	about.tsx	App name, student, registration, code, version, data-use statement. Links labelled as needing internet.

Justified differences

- **Welcome screen.** A short launch screen (name and Continue) plays before the tabs. It does not replace Home. The required sentence is still the first sentence of the portfolio.
- **Tab wording.** The dock says Home, About me, Work, Edit and Info. The brief’s section names (skills, projects, profile, about) are the screen titles and this document’s headings. The mapping is on page 2.
- **Floating dock.** The tab bar is a frosted capsule rather than a full-width system bar, so card content can scroll underneath with clearance. Labels and roles are unchanged.
- **Colour.** Early structure sketches were plain blocks. The built theme uses the pastel wash described on page 1. Hierarchy, contrast and the reading order did not change.
- **Screenshots.** These pages are the annotated wireframes of the submitted screens. The continuous demonstration video is the record of the installed APK.

Portfolio wording is personal: seven skills, education at INES-Ruhengeri and King David Academy, and two projects (TetaPortfolio and MilkCheck). Colours, the avatar and the app name were chosen for this submission.